Perineural Spread

A higher incidence of perineural spread has been reported in some studies, with Leibovitch et al. noting involvement in more than 50 percent of periocular basal cell carcinomas (BCCs), often leading to orbital invasion. These aggressive tumors frequently require extensive surgical intervention, including orbital exenteration in severe cases. Perineural spread may present clinically with pain, paresthesias, muscle weakness, or paralysis. The presence of focal neurologic symptoms at the site of a previously treated skin cancer should raise suspicion for nerve involvement.

Metastasis

Metastasis of BCC is rare, occurring in only 0.0028 to 0.55 percent of cases. Lymph node and lung involvement are the most common sites of metastatic spread. In a series by Von Domarus et al., three out of five patients with metastatic BCC had histologic evidence of perineural or intravascular invasion. Squamous differentiation was not seen in the primary tumors but was present in two of the five metastatic cases. Overall, squamous differentiation was observed in 15 percent of primary or metastatic tumors among 170 reviewed cases. Aggressive histologic features—such as morpheaform growth pattern, squamous metaplasia, and perineural invasion—are recognized risk factors for metastasis.

Diagnosis

Diagnosis of BCC is based on accurate histopathologic interpretation of skin biopsy specimens. Shave biopsy is often sufficient and preferred; a sterilized razor blade allows precise control over biopsy depth and is frequently superior to a No. 15 scalpel for this technique. Punch biopsy is particularly useful for evaluating flat lesions, such as morpheaform BCC, or recurrent tumors arising within a scar.

Histopathology

Histopathologic features vary by subtype, but most BCCs share common characteristics. The malignant basaloid cells typically have large nuclei with scant cytoplasm. Despite nuclear enlargement, overt nuclear atypia may be minimal, and mitotic figures are usually rare or absent. A hallmark feature is stromal retraction around tumor nests, creating peritumoral lacunae, which aids in diagnosis.

Nodular Basal Cell Carcinoma

Nodular BCC is the most common subtype, accounting for approximately 50 percent of cases. It is characterized by well-circumscribed nodules of basophilic tumor cells and prominent stromal retraction (Fig. 115-8A). Micronodular BCC refers to tumors composed of multiple small nodules less than 15 µm in diameter (Fig. 115-8B), which may have a higher risk of recurrence due to less distinct margins.

Pigmented Basal Cell Carcinoma

Pigmented BCC shares histologic features with nodular BCC but includes melanin deposition. Melanocytes are found interspersed among tumor cells and contain abundant melanin granules in their cytoplasm and dendrites. While tumor cells themselves contain little melanin, the surrounding stroma often harbors numerous melanophages. Although about 75 percent of BCCs contain melanocytes, significant pigmentation is present in only about 25 percent of cases.

Superficial Basal Cell Carcinoma

Superficial BCC is characterized by small buds of malignant basaloid cells extending from the epidermal basal layer into the upper dermis. The peripheral cells typically show palisading, and epidermal atrophy may be present. Dermal invasion is usually minimal. This subtype is most commonly found on the trunk and extremities but can also occur on the head and neck.